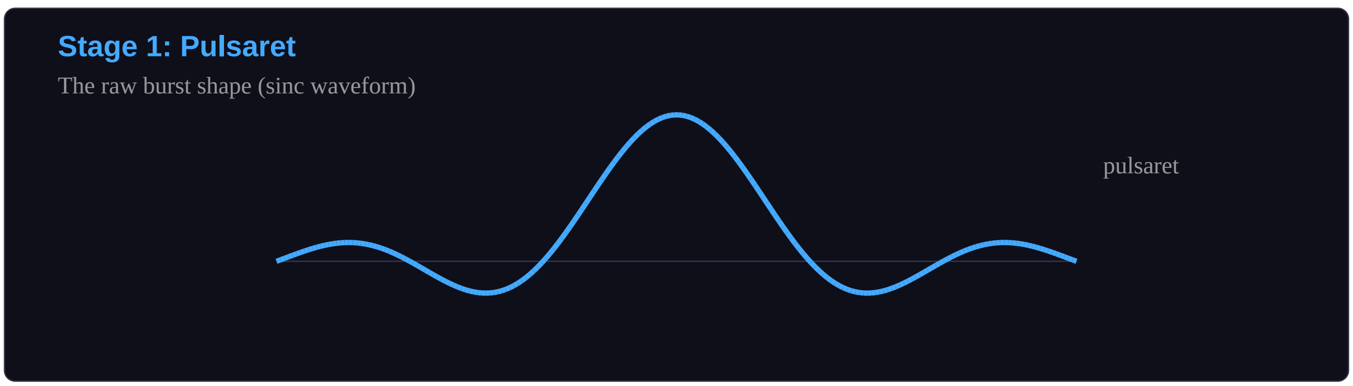
Spaluter

Pulsar Synthesis for disting NT

A pulsar synthesis instrument plugin for the Expert Sleepers disting NT Eurorack module. 68 parameters, 18 CV inputs, 4-voice polyphony, 10 pulsaret waveforms, 3 independent formants, stochastic and burst masking.



A sinc pulsaret — the raw waveform burst at the core of pulsar synthesis.

github.com/sroons/spaluter

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About Spaluter

Spaluter is a pulsar synthesis instrument for the Expert Sleepers disting NT. It generates sound by repeating short waveform bursts (pulsarets) at audio rates, producing timbres that range from thick analog-style bass and shimmering pads to metallic percussion, vowel-like vocal textures, granular noise clouds, and deep evolving drones. The synthesis engine provides direct control over spectral peaks (formants), pulse density (duty cycle), and micro-level stochastic processes (masking, jitter, glisson) that are difficult or impossible to achieve with conventional subtractive or FM synthesis.

With up to 4 polyphonic voices, 3 independent formant oscillators, 18 CV inputs (12 pre-routed by default), and real-time waveform display, Spaluter is designed for both immediate sonic exploration and deep, evolving patches under voltage control.

What Is Pulsar Synthesis?

TL;DR

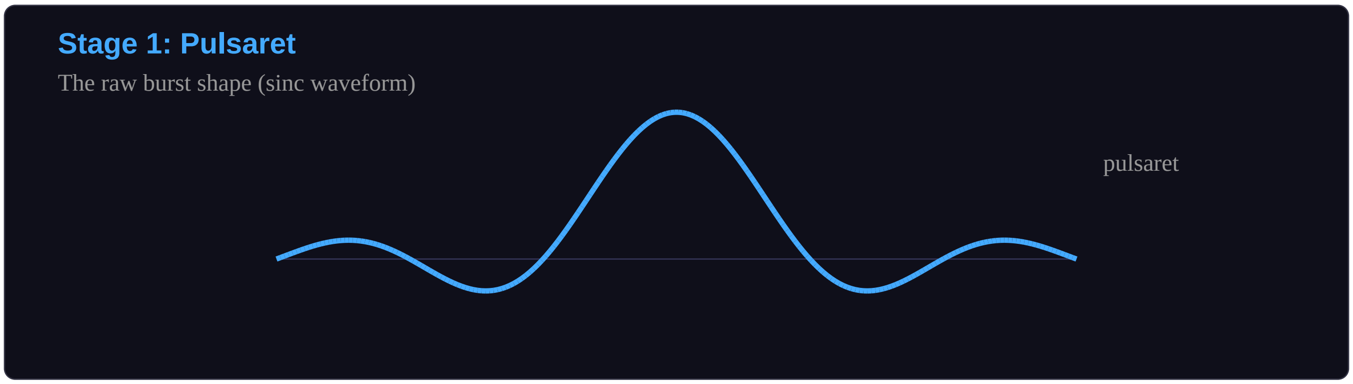
- A **pulsaret** (short waveform burst) is shaped by a **window** (amplitude envelope) and repeated at the **fundamental frequency** to produce a pitched tone.
- The **duty cycle** controls what fraction of each period is active sound vs. silence, dramatically affecting timbre.
- **Formants** (1-3 independent spectral peaks) cycle the pulsaret waveform at fixed frequencies, producing vowel-like resonances.
- **Masking** selectively mutes pulses for rhythmic or stochastic textures.

Pulsar synthesis is a technique developed by Curtis Roads and Alberto de Campo in the early 2000s, described in Roads' book *Microsound* (MIT Press, 2001). It belongs to the family of granular and particle-based synthesis methods that operate on the micro-timescale of sound — durations below approximately 100 milliseconds, where individual sonic events blur into continuous tones and textures.

The following five stages illustrate how a pulsar tone is constructed, matching the interactive explainer at the Spaluter project page.

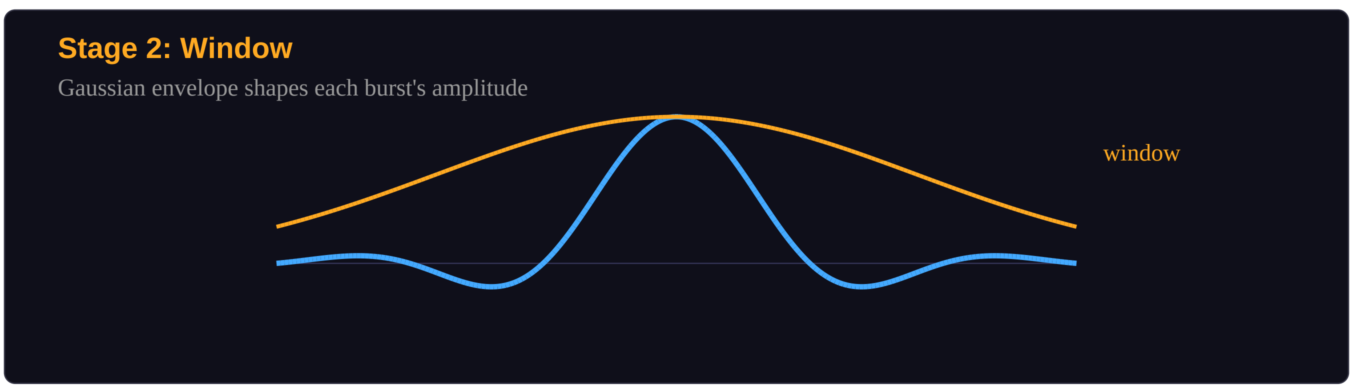
1. Pulsaret — the raw burst shape

A pulsaret is the waveform contained within a single pulse. It determines the basic harmonic content of the sound. Common pulsaret shapes include sine waves, sinc functions (band-limited impulses), and noise.



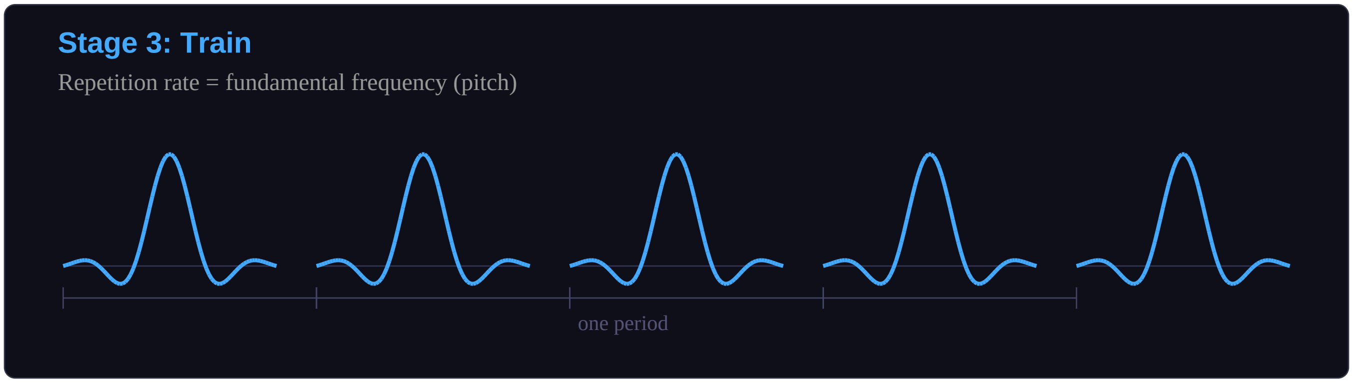
2. Window — the amplitude envelope per burst

A window function shapes the amplitude of each pulsaret, controlling how the burst fades in and out. The window determines the spectral smoothness of each pulse: a Gaussian window produces smooth, low-sidelobe spectra; a rectangular window maximizes brightness at the cost of spectral splatter.



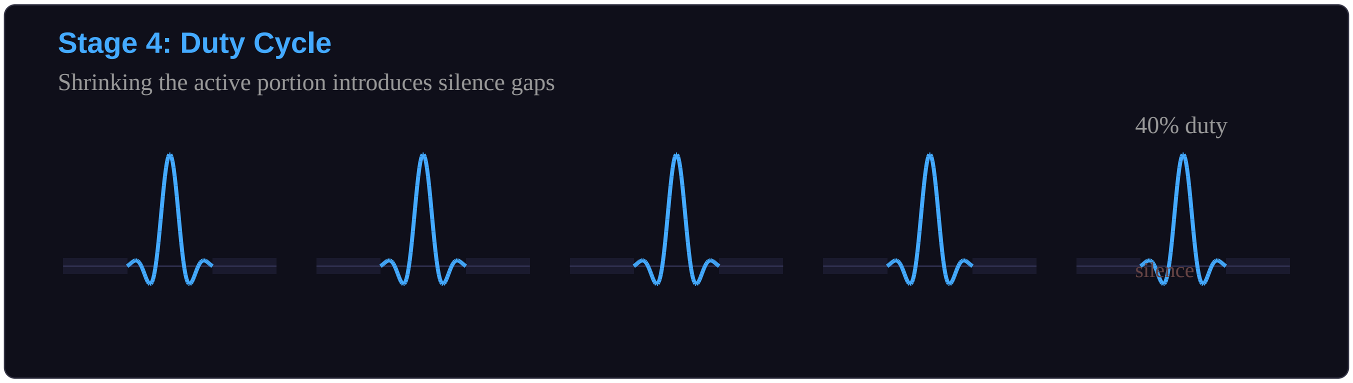
3. Train — repetition at the fundamental frequency

The windowed pulsaret is repeated periodically. The repetition rate determines the fundamental frequency (pitch) of the resulting tone. At audio rates (roughly 20 Hz and above), the train is perceived as a continuous pitched sound.



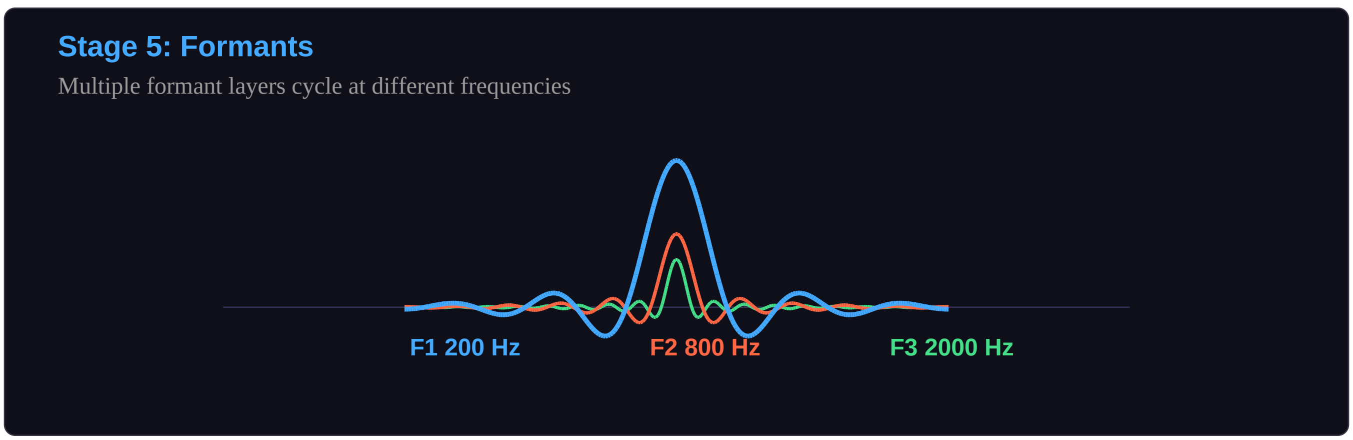
4. Duty Cycle — active portion vs. silence

The duty cycle specifies what fraction of each fundamental period contains active sound. The pulsaret is centered within the period with equal silence gaps before and after. At 100%% duty, the sound approaches conventional wavetable synthesis. As duty decreases, the characteristic hollow, buzzing quality of pulsar synthesis emerges. At very low duty cycles, individual pulsarets become audible as distinct sonic particles.



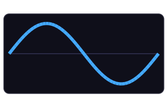
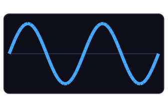
5. Formants — independent spectral peaks

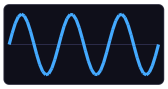
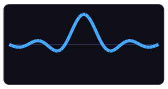
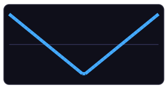
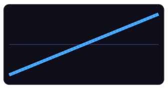
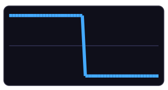
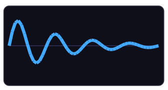
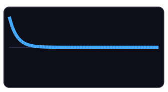
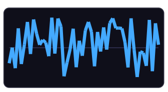
Formant oscillators cycle the pulsaret waveform at frequencies independent of the fundamental, creating fixed spectral peaks analogous to vocal formants in speech. By setting formant frequencies to values such as 200 Hz, 800 Hz, and 2000 Hz, vowel-like resonances emerge that remain constant as pitch changes. Spaluter provides up to 3 independent formant oscillators, each with its own frequency, pan position, and CV input.



Pulsaret Waveforms

Spaluter provides 10 pulsaret waveforms with continuous morphing between adjacent shapes. The Pulsaret parameter (0.0–9.0) selects and blends waveforms via bilinear interpolation between adjacent tables of 2048 samples each.

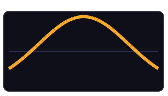
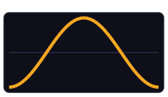
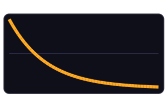
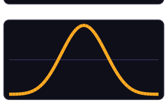
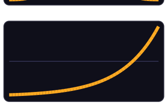
	**0.0 — Sine**	Pure fundamental. Clean starting point for isolating formant effects.
	**1.0 — Sine x2**	Second harmonic. Adds brightness without harshness.

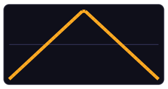
	**2.0 — Sine x3**	Third harmonic. Fuller harmonic content.
	**3.0 — Sinc**	Band-limited impulse with natural sidelobes. The classic pulsar synthesis waveform, closest to Roads' original work.
	**4.0 — Triangle**	Odd harmonics with rapid rolloff. Mellow, clarinet-adjacent timbre.
	**5.0 — Saw**	All harmonics present. Familiar subtractive-style brightness.
	**6.0 — Square**	Odd harmonics, hollow quality. Clarinet-like character.
	**7.0 — Formant**	Built-in resonant peak. Stacking with formant frequency parameters creates double-resonance effects.
	**8.0 — Pulse**	Extremely bright and nasal. Sharp transient with rapid decay.
	**9.0 — Noise**	Stochastic waveform. Replaces pitched content with noise bursts for percussion or breathy textures.

Fractional morph values (e.g. 3.5) blend between adjacent shapes via linear interpolation.

Window Functions

The window function shapes the amplitude envelope applied to each pulsaret. The Window parameter (0.0–8.0) morphs between 9 window types.

	**0.0 — Rectangular**	No fade — hard edges. Maximum brightness and click. Highest spectral energy but also highest sidelobe levels.
	**1.0 — Gaussian**	Smooth bell curve with sigma=0.3. Natural fade-in and fade-out with strong sidelobe suppression.
	**2.0 — Hann**	Raised cosine: 0.5 * (1 - cos(2*pi*p)). Smooth endpoints and a broad rounded peak.
	**3.0 — Exp Decay**	Sharp attack with exponential ringout. Percussive, plucked character. Pairs well with sinc pulsaret.
	**4.0 — Linear Decay**	Sharp attack with linear ramp down. Simpler percussive shape than exponential.
	**5.0 — Tukey**	Tapered cosine with alpha=0.5. Smooth fade-in/out edges with a flat center plateau.
	**6.0 — Blackman-Harris**	Four-term Blackman-Harris curve. Very low sidelobes and very smooth spectral edges.
	**7.0 — Reverse Exp**	Slow exponential swell into a sharp cutoff. Reverse-plucked or bowed-style articulation.



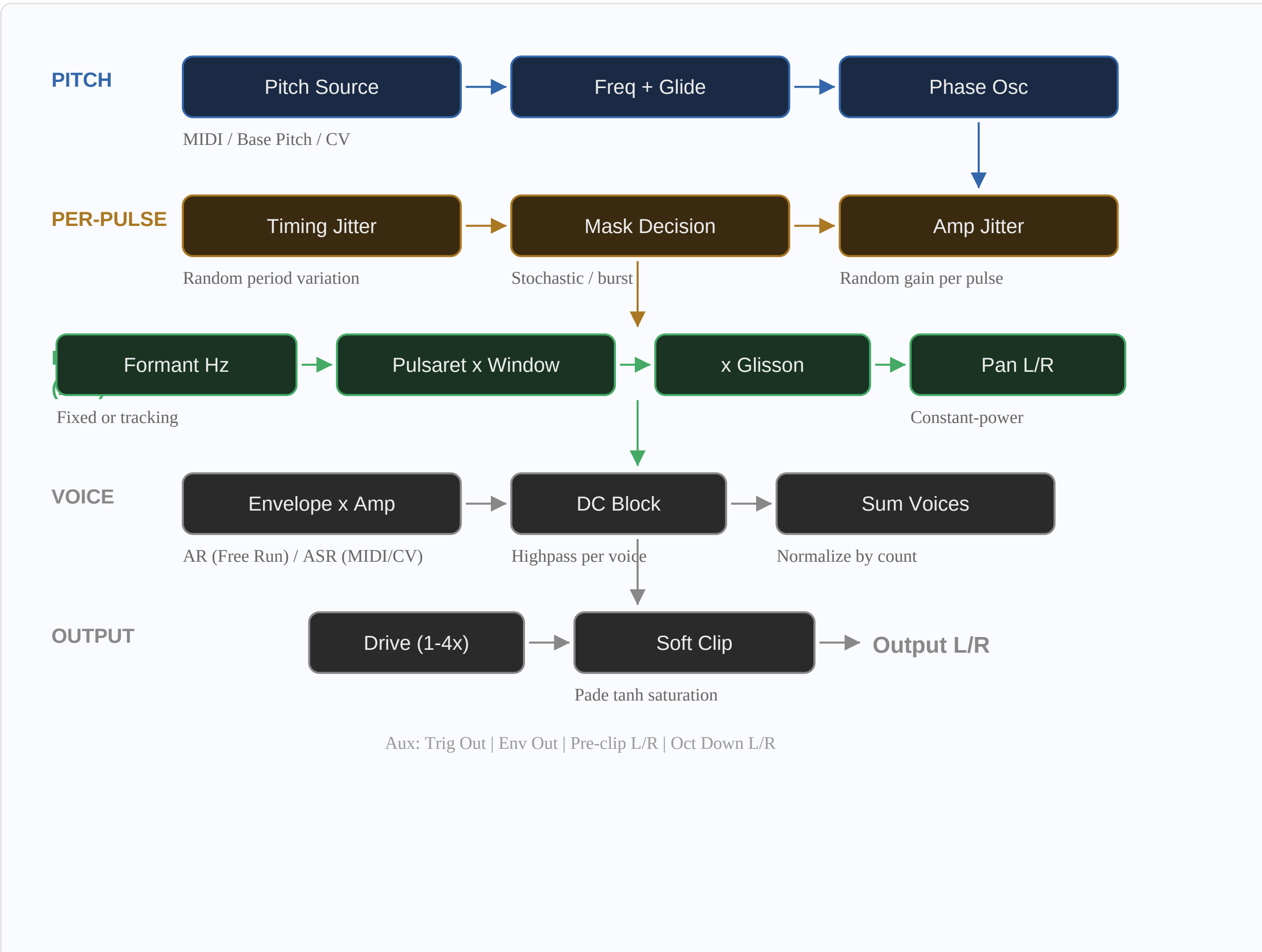
8.0 — Triangle

Linear rise to the midpoint, then linear fall. Symmetric ramp with a pointed peak.

Fractional values blend adjacent windows. The window interacts with glisson: exponential decay makes pitch sweeps audible mainly at the attack.

Signal Chain

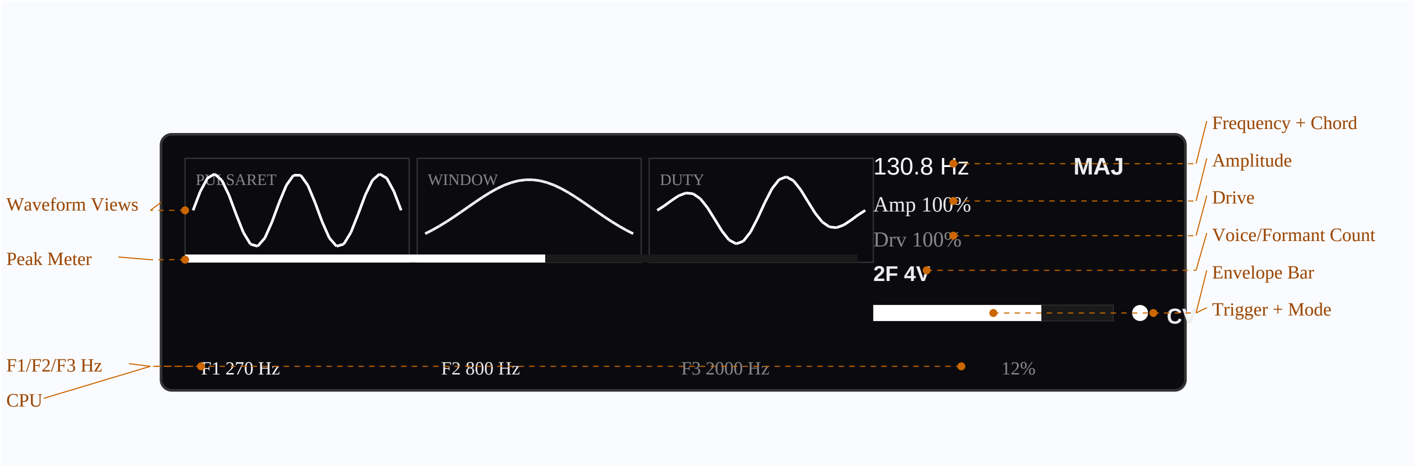
The signal chain flows from pitch source through per-pulse processing, formant synthesis, voice envelope, and final output stage. Each voice (1–4) runs the full chain independently; voices are summed and normalized before the drive and soft clip stages.



Blue = pitch section, amber = per-pulse decisions, green = formant loop (runs 1–3 times per pulse), gray = voice and output mixing.

The Display

The 256x64 pixel OLED display provides real-time visual feedback across three waveform previews and a synthesis state readout area. All waveform views respond to CV modulation in real time.



Annotated display layout showing waveform views, formant readouts, envelope bar, gate indicator, mode label, and peak meter.

Pot Mappings

Pot	Parameter	Range
Pot L	Pulsaret morph	0.0–9.0 (sweeps all 10 waveforms)
Pot L + press	Formant 1 Hz	20–2000 Hz (relative)
Pot L press only	Formant 1 On/Off	Toggle
Pot C	Window morph	0.0–8.0 (sweeps all 9 windows)
Pot C + press	Formant 2 Hz	20–2000 Hz (relative)
Pot C press only	Formant 2 On/Off	Toggle
Pot R	Duty Cycle	1–100%
Pot R + press	Formant 3 Hz	20–2000 Hz (relative)
Pot R press only	Formant 3 On/Off	Toggle

How to Use Spaluter

Getting Sound Immediately

Spaluter defaults to **CV mode** with **Amplitude at 100%%**. To produce sound, patch a trigger/gate signal into **Input 2** (Trigger CV) and a 1V/oct pitch source into **Input 1** (Pitch CV). Each rising edge on the trigger input allocates a new voice at the current pitch. The remaining 10 default CV inputs add modulation when patched.

Alternatively, switch **Gate Mode** to **Free Run** on the Mode page for immediate sound without any CV patching. Free Run generates a continuous tone at the Base Pitch frequency (default C1, ~32.7 Hz).

Modifying Timbre

- **Pulsaret shape** (Pot L or Waveform page): Sweeps through 10 waveform types. Start with Sinc (3.0) for the classic pulsar sound, then explore Sine (0.0) for purity or Noise (9.0) for texture.
- **Window function** (Pot C or Waveform page): Shapes the attack and decay of each burst. Gaussian (1.0) for smooth tones, Rectangular (0.0) for maximum brightness, Exp Decay (3.0) for plucked character.
- **Duty cycle** (Pot R or Waveform page): The most dramatic timbral control after formant frequency. 80–100%% is full and warm; 30–60%% is the characteristic pulsar buzz; 5–20%% produces sparse particle textures.
- **Formant frequencies** (Formants page): Set spectral peak locations. F1=270, F2=1090 produces an "ah" vowel. F1=200, F2=347, F3=511 produces metallic bell tones. Modulate with CV for expressive sweeps.

Parameter Interactions

Duty cycle and formant perception: As duty cycle decreases, the pulsaret occupies less of each period. This narrows the spectral envelope of each pulse, making formant peaks less distinct at very low duty values. At high duty, formants are clearly defined. Formant Duty Mode automates this relationship by tying duty to formant frequency.

Formant tracking: With Formant Track set to Fixed (default), formant frequencies remain constant regardless of pitch — producing the classic vocal formant effect where timbre changes with pitch. With Track enabled, formant frequencies scale proportionally with voice pitch, preserving the spectral shape like a sampler would.

Masking + jitter for organic textures: Stochastic masking (random pulse muting) combined with amplitude jitter (random gain variation) and timing jitter (random period variation) layers three independent sources of randomness. Low amounts (10–25%% each) produce subtle organic variation; higher amounts dissolve the tone into a granular particle cloud. With Independent Masking enabled, each formant receives its own mask decision, creating spectral flickering.

Navigating Parameters

disting NT Parameter Navigation

Spaluter's 68 parameters are organized across 15 pages. On the disting NT hardware, use the **left encoder** to scroll through parameter pages and the **right encoder** to adjust the selected parameter value. Press an encoder button to toggle between page selection and value editing modes.

Spaluter maps **3 pots** (Pulsaret, Window, Duty) and **4 buttons** (mask mode, formant count, voice count, chord type) for direct, page-free control of the most frequently adjusted parameters.

NT Helper

NT Helper is a companion app (available for iPad and as a browser-based tool) that connects to the disting NT over USB. It provides a graphical interface for editing all parameters simultaneously, making it faster to configure complex patches than scrolling through pages on the module's display. All parameter changes made in NT Helper are reflected on the module in real time, and vice versa.

Parameters

68 parameters across 15 pages:

Page	Parameter	Range	Default
Mode	Gate Mode	MIDI / Free Run / CV	CV
	Voice Count	1–4	1
	Chord Type	14 types	Unison
	MIDI Ch	1–16	1
	Base Pitch	MIDI 0–127	C1 (24)
Level	Amplitude	0–200%	100%
	Drive	100–400%	100%
	Attack	0.1–2000 ms	10 ms
	Release	1.0–3200 ms	200 ms
	Glide	0–2000 ms	0 ms
Waveform	Pulsaret	0.0–9.0	2.5
	Window	0.0–8.0	0.5
	Duty Cycle	1–100%	50%
	Duty Mode	Manual / Formant	Manual
Formants	Formant Count	1–3	2
	Formant 1 Hz	20–2000 Hz	20 Hz
	Formant 1 On	Off / On	On
	Formant 2 Hz	20–2000 Hz	200 Hz
	Formant 2 On	Off / On	On
	Formant 3 Hz	20–2000 Hz	400 Hz

Page	Parameter	Range	Default
	Formant 3 On	Off / On	On
	Formant Track	Fixed / Track	Fixed
Texture	Mask Mode	Off / Stochastic / Burst	Off
	Mask Amount	0–100%	50%
	Burst On	1–16	4
	Burst Off	0–16	4
	Indep Mask	Off / On	Off
Texture	Amp Jitter	0–100%	0%
	Time Jitter	0–100%	0%
	Glisson	-10.0 to +10.0	0
Panning	Pan 1	-100 to +100	0
	Pan 2	-100 to +100	-50
	Pan 3	-100 to +100	+50
Sample	Use Sample	Off / On	Off
	Folder	(SD card)	—
	File	(SD card)	—
	Sample Rate	25–400%	100%
Outputs	Output L	Bus 1–64	Bus 13
	Output L mode	Add / Replace	Add
	Output R	Bus 1–64	Bus 14
	Output R mode	Add / Replace	Add
Aux Out	Trig Out	Bus 0–64	0 (none)
	Trig Out mode	Add / Replace	Add
	Env Out	Bus 0–64	0 (none)
	Env Out mode	Add / Replace	Add
	Pre-clip L	Bus 0–64	0 (none)
	Pre-clip L mode	Add / Replace	Add
	Pre-clip R	Bus 0–64	0 (none)
	Pre-clip R mode	Add / Replace	Add
	Oct Down L	Bus 0–64	0 (none)
	Oct Down L mode	Add / Replace	Add
	Oct Down R	Bus 0–64	0 (none)

Page	Parameter	Range	Default
	Oct Down R mode	Add / Replace	Add

Unused parameters are automatically grayed out based on context (e.g., Chord Type in MIDI/CV mode, Burst params when mask mode is not Burst).

CV Inputs

All CV inputs are **bipolar** (±5V). Each is routable to any of the 64 buses (1–12 inputs, 13–20 outputs, 21–64 aux), or 0 for none. CV modulation is applied as an offset on top of the parameter's base value. All CVs except Pitch are block-rate averaged. Pitch CV is processed per-sample for accurate 1V/oct tracking.

CV Input	Default Bus	Scaling	Effect
Pitch CV	Input 1	1V/oct exponential	Per-sample pitch modulation
Trigger CV	Input 2	>2.5V = high	Trigger for CV mode voice allocation
Duty CV	Input 3	±5V → ±20%	Duty cycle offset added to base
Mask CV	Input 4	±5V → ±50%	Mask amount offset (bipolar)
Pulsaret CV	Input 5	±5V → full range	Pulsaret morph ±4.5
Window CV	Input 6	±5V → full range	Window morph ±4.0
Formant 1 CV	Input 7	±5V → ±1000 Hz	Formant 1 frequency offset
Formant 2 CV	Input 8	±5V → ±1000 Hz	Formant 2 frequency offset
Formant 3 CV	Input 9	±5V → ±1000 Hz	Formant 3 frequency offset
Attack CV	Input 10	±5V → ±1000 ms	Envelope attack time offset
Release CV	Input 11	±5V → ±1600 ms	Envelope release time offset
Glisson CV	Input 12	±5V → ±2.0 oct	Glisson depth offset
Amplitude CV	0 (none)	±5V → ±50%	Amplitude offset added to base
Pan 1 CV	0 (none)	±5V → ±100%	Formant 1 stereo pan position
Pan 2 CV	0 (none)	±5V → ±100%	Formant 2 stereo pan position
Pan 3 CV	0 (none)	±5V → ±100%	Formant 3 stereo pan position
Amp Jit CV	0 (none)	±5V → ±50%	Amp jitter amount offset
Time Jit CV	0 (none)	±5V → ±50%	Timing jitter amount offset

CV Mode (Rings-style Voice Triggering)

Polyphonic voice triggering from a single trigger+pitch CV pair, inspired by Mutable Instruments Rings. Each trigger rising edge allocates a new voice while previous voices ring out through their release envelopes — up to 4 simultaneous voices.

Setup

1. Set **Gate Mode** to **CV** (Mode page) — this is the default.
2. Patch a trigger/gate signal into **Input 2** (Trigger CV, default bus).
3. Patch a 1V/oct pitch source into **Input 1** (Pitch CV, default bus).
4. Adjust **Attack** and **Release** on the Level page. Amplitude defaults to 100%%.

Behavior

Trigger State	What Happens
Rising edge	Allocates a voice, captures pitch from Base Pitch x Pitch CV, starts attack
Held high	Active voice tracks Pitch CV in real time (enables pitch bends)
Falling edge	Active voice enters release; pitch and all synthesis parameters freeze
During release	Voice sounds independently — knob/CV changes only affect the next triggered voice

When all 4 voices are releasing, a new trigger steals the quietest (lowest envelope) voice. Voice Count, Chord Type, and MIDI Ch are not used in CV mode and are automatically grayed out.

Hardware Controls

These mappings are active on the realtime display screen. Encoders and buttons not listed below retain their standard disting NT behavior.

Pots

Pot	Parameter	Range
Pot L	Pulsaret morph	0.0–9.0 (sweeps all 10 waveforms)
Pot L + press	Formant 1 Hz	20–2000 Hz (relative)
Pot L press only	Formant 1 On/Off	Toggle
Pot C	Window morph	0.0–8.0 (sweeps all 9 windows)
Pot C + press	Formant 2 Hz	20–2000 Hz (relative)
Pot C press only	Formant 2 On/Off	Toggle
Pot R	Duty Cycle	1–100%
Pot R + press	Formant 3 Hz	20–2000 Hz (relative)
Pot R press only	Formant 3 On/Off	Toggle

Buttons

Button	Action
Left encoder button	Cycle mask mode: Off → Stochastic → Burst → Off
Right encoder button	Cycle formant count: 1 → 2 → 3 → 1
Button 3	Cycle voice count: 1 → 2 → 3 → 4 → 1
Button 4	Cycle chord type (14 options)

Outputs

Output L and R are routable to any bus via the Outputs page (default: L = bus 13, R = bus 14). Six optional auxiliary outputs are available on the Aux Out page, all defaulting to bus 0 (disabled):

Output	Signal	Use
Trig Out	1.0 on each new pulse (voice 0)	Clock/trigger synced to pulse rate
Env Out	Max envelope across all voices	Envelope follower for external modulation
Pre-clip L/R	Stereo before soft clip	Raw signal for external processing
Oct Down L/R	Stereo frequency divider	Sub-octave, one octave below fundamental

Sound Design Tips

Formant Frequencies

Formant frequencies have the most dramatic effect on timbre. They define spectral peaks analogous to vocal formants in speech.

- **Low formants (20–100 Hz)**: Deep, subharmonic rumbles. F1=20 Hz produces a thick bass drone.
- **Vowel-like tones**: F1=270, F2=1090 for "ah"; F1=300, F2=870 for "oo." Experiment with vocal formant charts.
- **Harmonic stacking**: Set formants to integer multiples of the fundamental for bright, reinforced harmonics.
- **Inharmonic/metallic**: Non-integer ratios (F1=200, F2=347, F3=511) produce bell-like or metallic timbres.
- **Formant Track mode**: Scales formant frequencies with pitch. A formant at 400 Hz doubles to 800 Hz one octave up, preserving spectral shape across the keyboard.

Duty Cycle

- **80–100%%**: Full, warm — approaching wavetable synthesis.
- **30–60%%**: The characteristic pulsar "buzz." Sweet spot for most patches.
- **5–20%%**: Sparse, clicking, particle-like textures. Amp jitter is especially audible here.
- **Formant Duty Mode**: Automatically ties duty to formant frequency, keeping pulsaret waveform cycles consistent.

Masking

- **Stochastic masking** at 10–30%% adds subtle grit. At 70–90%% the tone dissolves into a sparse particle cloud.
- **Burst masking** creates micro-rhythmic patterns. Try On=1, Off=3 for stutter; On=7, Off=1 for subtle hiccup.
- **Indep Mask** gives each formant its own mask decision. In stochastic mode, different formant subsets sound on each pulse. In burst mode, formants alternate.
- **Amp jitter + stochastic masking**: Two layers of randomness — some pulses quieter, some missing entirely.

Effects

- **Timing Jitter**: 5–15%% adds warmth and analog drift. With multiple unison voices, each drifts independently for natural chorusing.
- **Glisson**: Sweeps pitch within each pulsaret. Subtle values (±0.3–1.0) add shimmer; extreme values (±7–10) create laser-like chirps.
- **Amp Jitter**: 10–20%% adds organic variation like a bowed string. High values (60–100%%) make the sound fragile.

Polyphony and Chords

In Free Run mode, additional voices are tuned relative to the base pitch by the selected Chord Type. **Harmonic intervals** (Unison, Octaves, Fifths, Sub+Oct) use pure frequency ratios. **Tonal chords** (Major, Minor, Maj7, etc.) use equal temperament semitone offsets. Each voice has independent phase, envelope, masking, and jitter state. Volume is normalized by voice count parameter (not active voices) to prevent level jumps.

Patch Ideas

Starting points for exploring Spaluter. Each patch lists specific parameter settings and describes the resulting sound.

Bass Drone: Sinc (3.0), Gaussian, 40%% duty, F1=80 Hz, Free Run, base pitch C1. Slow LFO on Formant 1 CV for movement. A deep, slowly evolving low-frequency drone.

Vocal Pad: Sine (0.0), Hann, 60%% duty, F1=270, F2=1090, F3=800. 4 voices Maj7. Slow LFO on F1 CV to morph vowels. Lush choral pad with shifting vowel-like timbre.

Percussive Texture: Pulse (8.0), Exp Decay, 20%% duty, F1=400. Burst mask 2on/3off. Short attack, short release. Glisson +3.0. Rhythmic clicking bursts with pitch-swept transients.

Evolving Ambient: Sinc (3.0), Gaussian, 50%% duty, F1=200, F2=600, F3=1200. 4 voices Octaves. Stochastic mask 20%%, Amp Jit 15%%, Time Jit 10%%, Indep Mask on. Slow CV on all 3 formant inputs. A shimmering, constantly shifting ambient texture.

Metallic Bell: Triangle (4.0), Hann, 70%% duty, F1=200, F2=347, F3=511 (inharmonic ratios). CV mode, short attack, long release (2000 ms). No masking. Struck bell tone with non-harmonic partials and long, ringing decay.

Granular Cloud: Noise (9.0), Gaussian, 15%% duty, stochastic mask 60%%, Amp Jit 40%%, Time Jit 30%%. Free Run, 2 voices Unison. Very sparse, dissolving texture of random noise grains.

Organ Tone: Square (6.0), Rectangular, 100%% duty, F1=fundamental, Formant Track on. 4 voices Power chord. Classic hollow organ sound with stacked fifths.

Laser Chirps: Sinc (3.0), Exp Decay, 30%% duty, Glisson +8.0. Short attack, medium release. Each pulse sweeps pitch dramatically. Add CV trigger for rhythmic firing. Sci-fi laser and chirp effects.

Sub Bass Layer: Saw (5.0), Hann, 80%% duty, F1=40 Hz. Free Run, base pitch C0 (MIDI 12). Use sub-octave output routed to a separate bus for layering. Deep, warm sub-bass foundation.

Spectral Freeze: Sine x3 (2.0), Gaussian, 50%% duty, F1=300, F2=900. CV mode with long release (3000 ms). Each triggered voice freezes its parameters on release. Layer 4 frozen timbres by triggering at different settings for a spectral snapshot collage.

Further Reading

- Curtis Roads, *Microsound* (MIT Press, 2001) — the definitive reference on pulsar synthesis and micro-timescale sound.
- Curtis Roads, "Sound Composition with Pulsars," *Journal of the Audio Engineering Society*, 2001.
- Curtis Roads, *The Computer Music Tutorial* (MIT Press, 1996) — broader context on granular and particle synthesis.
- Alberto de Campo — related implementations and research on microsound techniques.
- Wikipedia: Pulsar Synthesis — en.wikipedia.org/wiki/Pulsar_synthesis

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github.com/sroons/spaluter